Question 1

```
#a
sal <- read.table("salamanders.txt")
sal
```

```
     V1    V2     V3   V4
1  pool abund fecund  CWD
2     1   0.0    0.0  4.7
3     2  23.6    5.3  7.7
4     3  12.6    4.8  5.1
5     4 108.1    6.0 12.1
6     5  16.9    6.7  3.1
7     6   5.6    0.0  2.4
8     7   0.9    0.1  2.4
9     8   0.2    0.0  2.2
10    9   3.2    1.6  2.6
11   10   0.5    0.0  2.7
12   11   1.1    0.4  2.7
13   12  28.6    4.1  2.7
14   13   0.8    0.0  2.5
15   14   0.1    0.0  2.0
```

```
View(sal)

#b
# structures
str(sal)
```

```
'data.frame':	15 obs. of  4 variables:
 $ V1: chr  "pool" "1" "2" "3" ...
 $ V2: chr  "abund" "0.0" "23.6" "12.6" ...
 $ V3: chr  "fecund" "0.0" "5.3" "4.8" ...
 $ V4: chr  "CWD" "4.7" "7.7" "5.1" ...
```

```
# number of rows
nrow(sal)
```

```
[1] 15
```

```
# number of columns/variables
ncol(sal)
```

```
[1] 4
```

```
# c
colnames(sal) <- sal[1, ]
```

```
sal <- sal[-1, ]


View(sal)
```

Question 3

```
# a

std_dev <- sd(sal$CWD)

std_dev
```

```
[1] 2.81976
```

```
avg <- mean(as.numeric(sal$CWD))
avg
```

```
[1] 3.921429
```

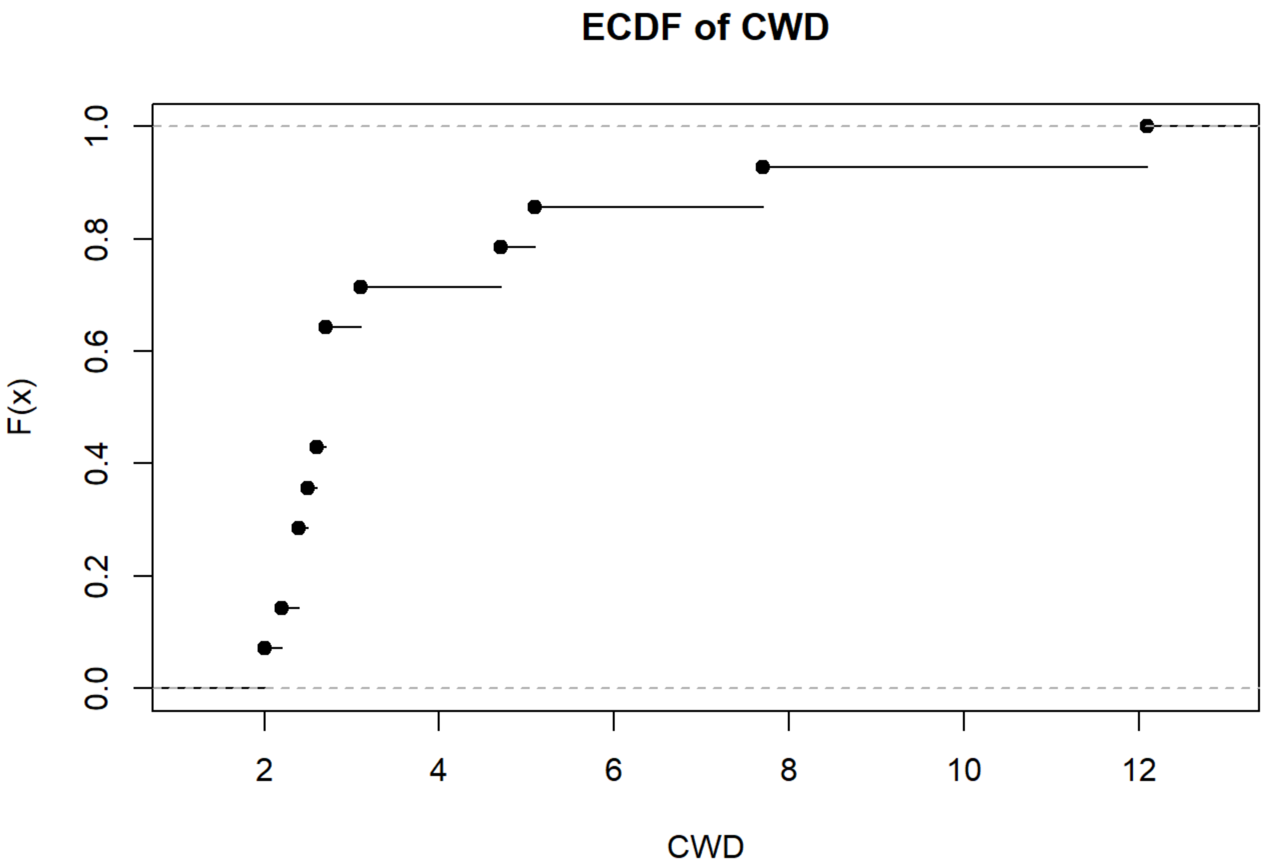
```
cv <- std_dev/avg *100

cv
```

```
[1] 71.90644
```

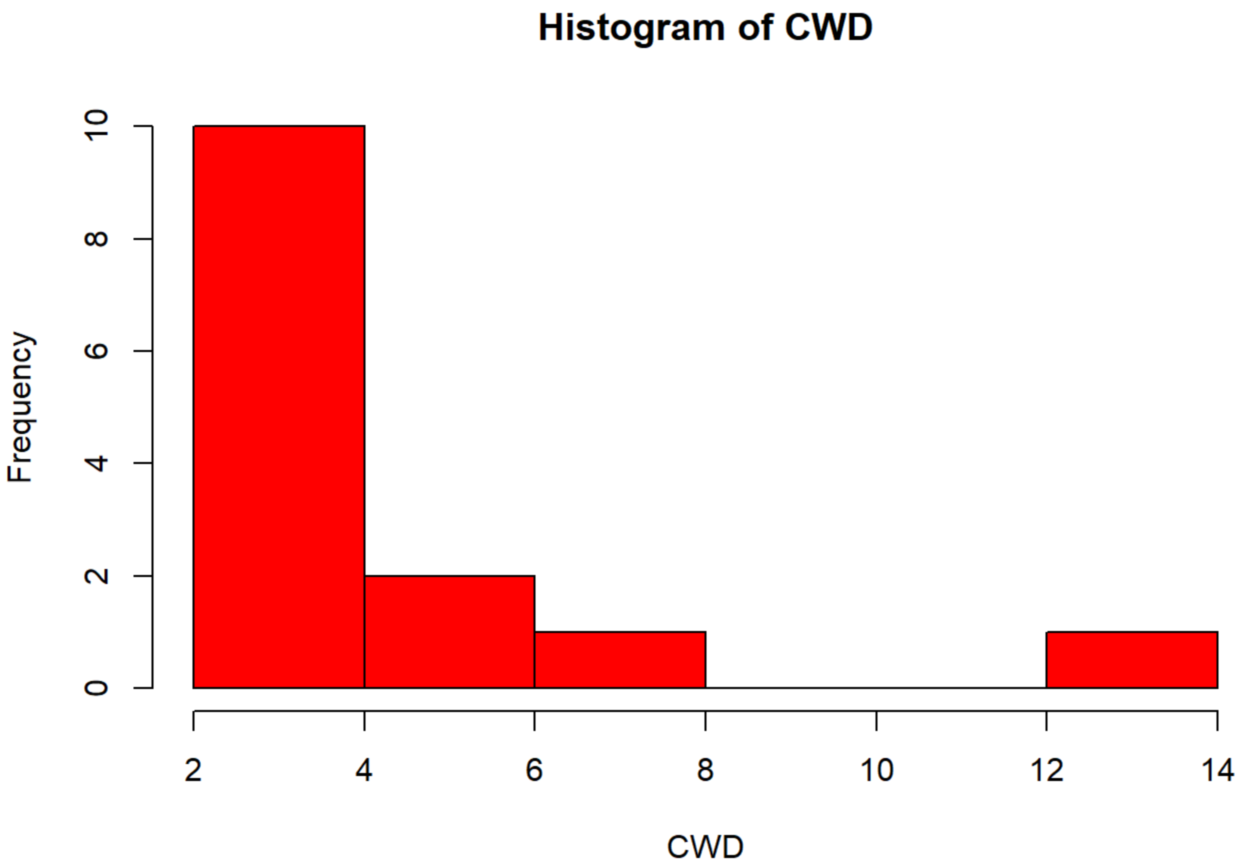
```
# b

iqr <- IQR(sal$CWD)
```

Question 4

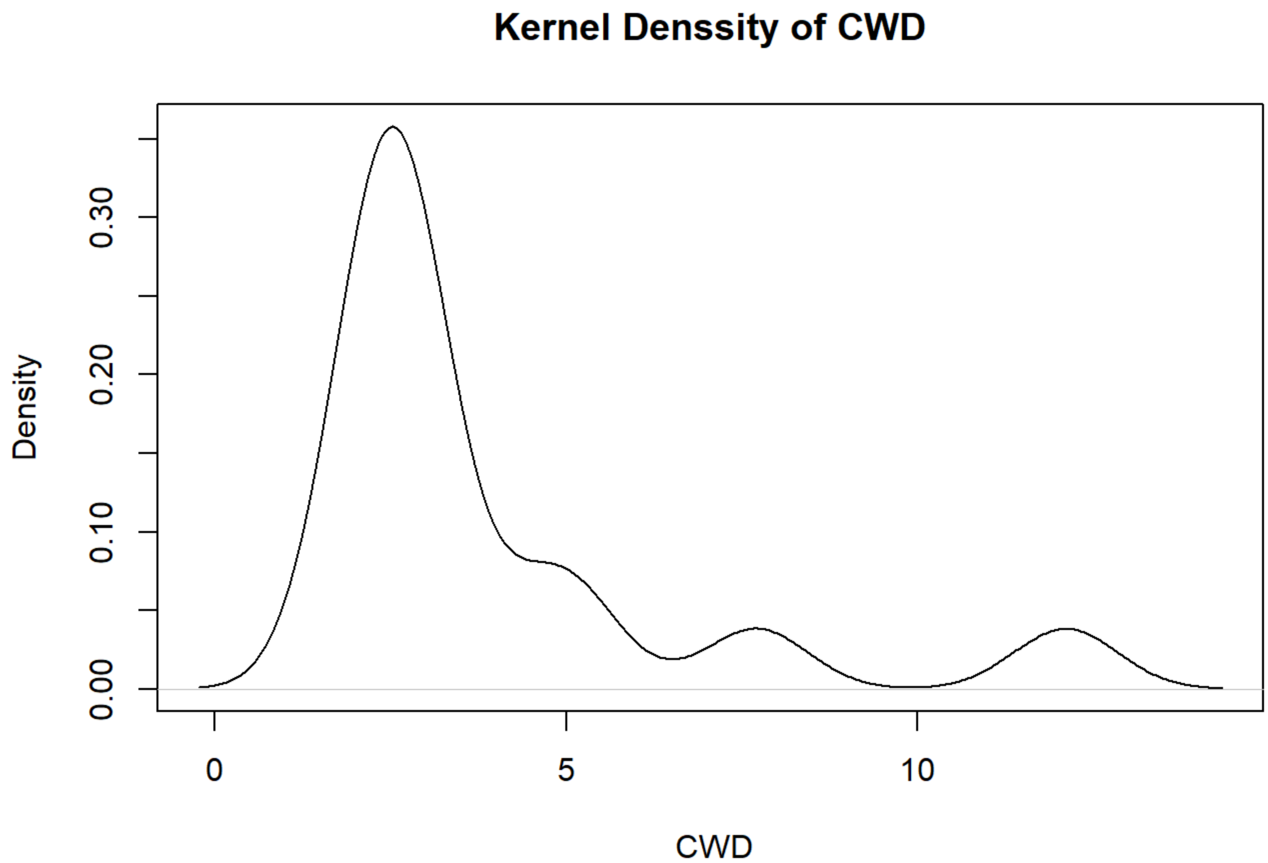
```
# a
# ecdf graph
plot(ecdf(as.numeric(sal$CWD)),
     main = "ECDF of CWD",
     xlab = "CWD",
     ylab = "F(x)")
```



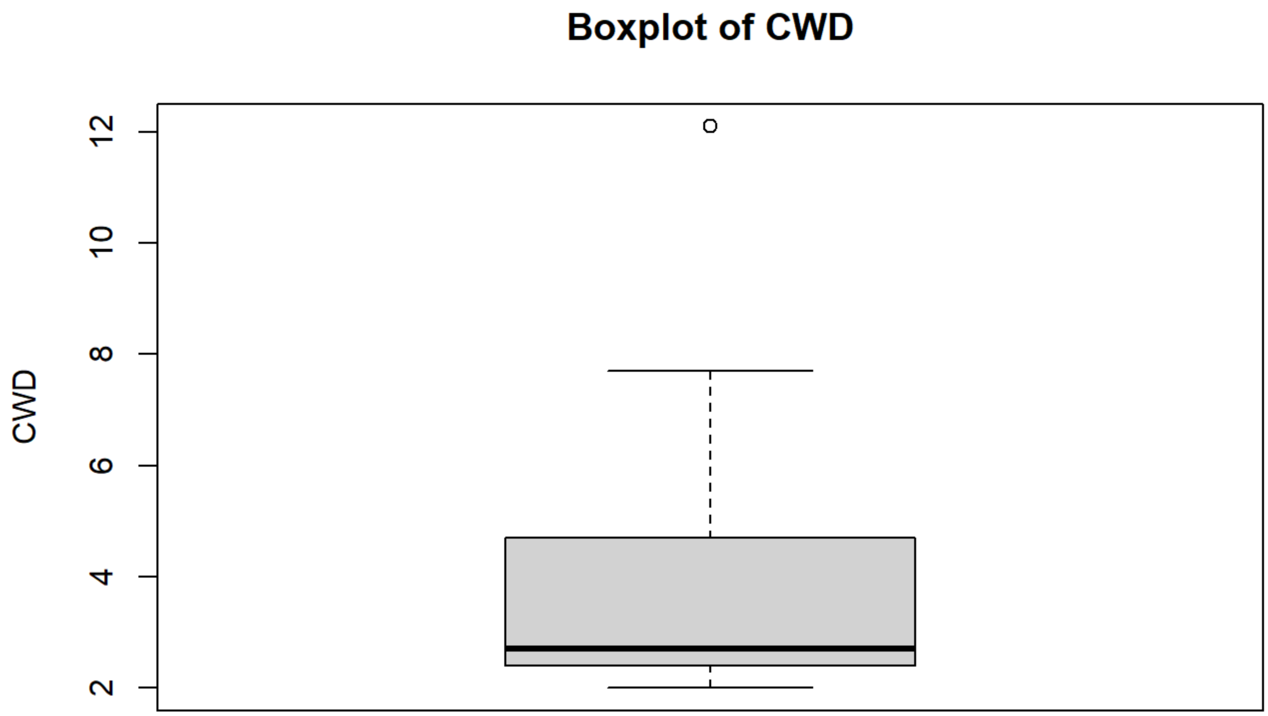

```
# b
# histogram
hist(as.numeric(sal$CWD),
     main = "Histogram of CWD",
     xlab = "CWD",
     col = "red")
```




```
# c
plot(density(as.numeric(sal$CWD)),
     main = "Kernel Denssity of CWD",
     xlab = " CWD")
```




```
# d
boxplot(as.numeric(sal$CWD),
        main = "Boxplot of CWD",
        ylab = "CWD")
```



Question 5

```
stat_vars <- sal[, c("abund", "fecund", "CWD")]

View(stat_vars)

stat_vars$abund <- as.numeric(stat_vars$abund)

stat_vars$fecund <- as.numeric(stat_vars$fecund)

stat_vars$CWD <- as.numeric(stat_vars$CWD)


# a
# covariance matrix
cov(stat_vars)
```

```
           abund    fecund       CWD
abund  817.62418 50.112088 70.889780
fecund  50.11209  7.037582  4.729890
CWD     70.88978  4.729890  7.951044
```

```r
# b
# pearson's correlation matrix
cor(stat_vars, method = "pearson")
```

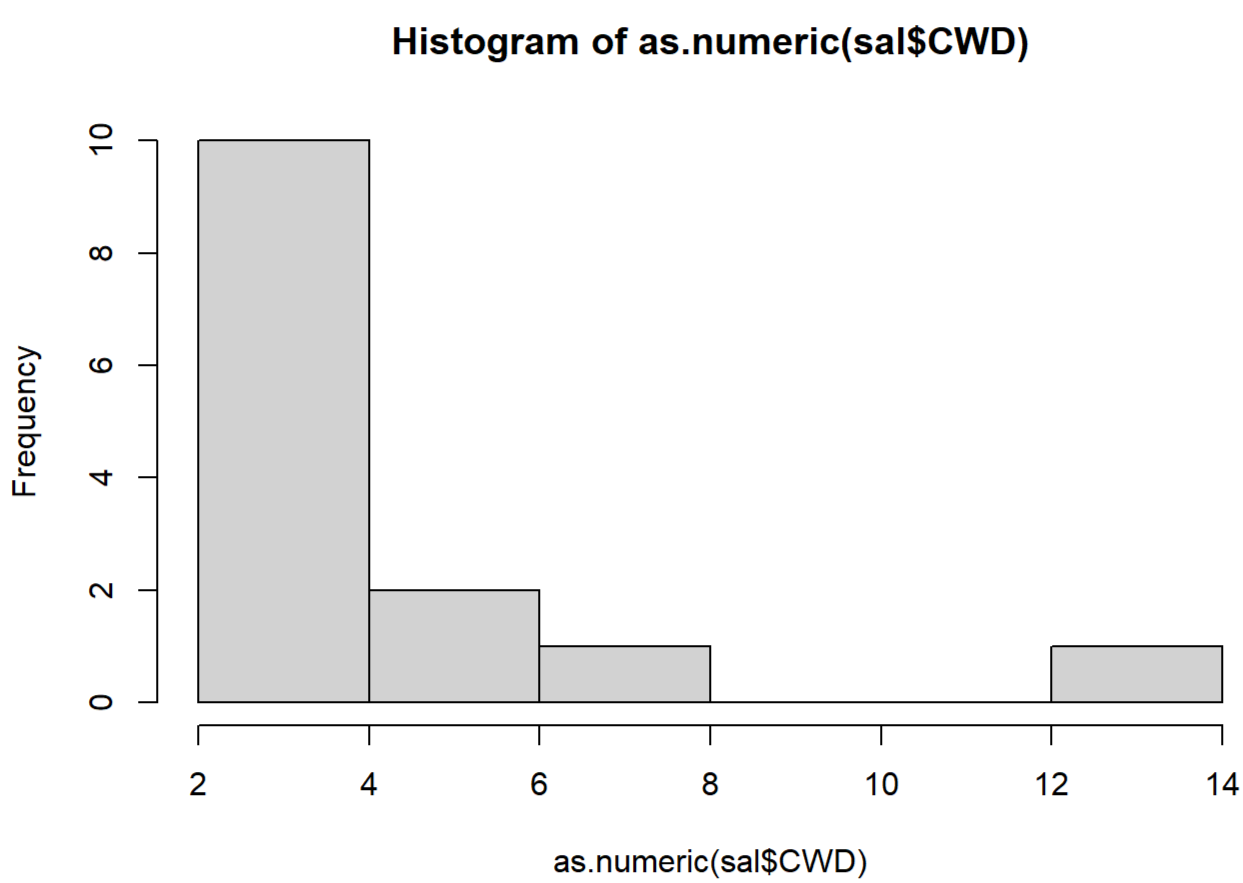
```
           abund    fecund       CWD
abund  1.0000000 0.6606233 0.8792142
fecund 0.6606233 1.0000000 0.6323059
CWD    0.8792142 0.6323059 1.0000000
```

```r
# c
# spearman correlation matrix
cor(stat_vars, method = "spearman")
```

```
           abund    fecund       CWD
abund  1.0000000 0.8532543 0.5790144
fecund 0.8532543 1.0000000 0.6981163
CWD    0.5790144 0.6981163 1.0000000
```

Question 6

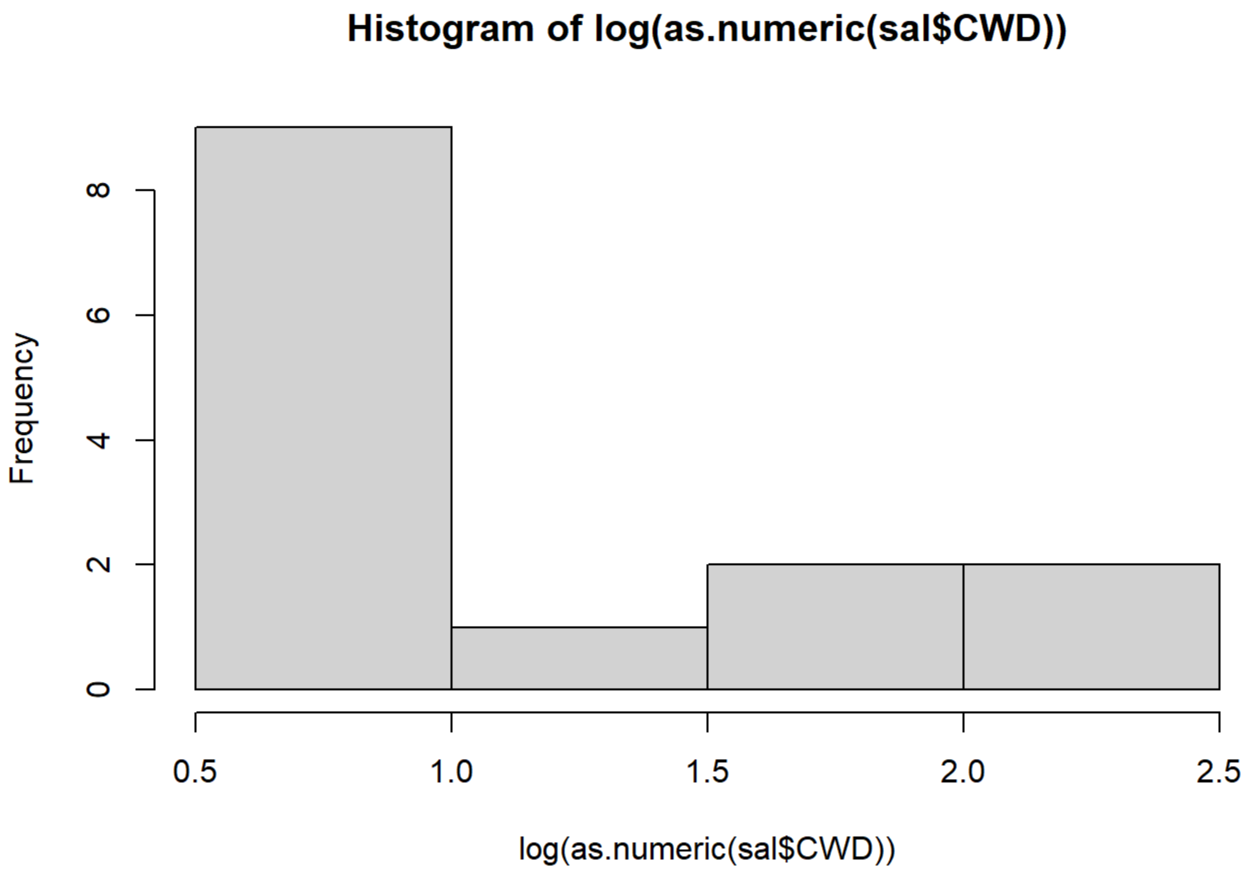
```r
hist(as.numeric(sal$CWD))
```



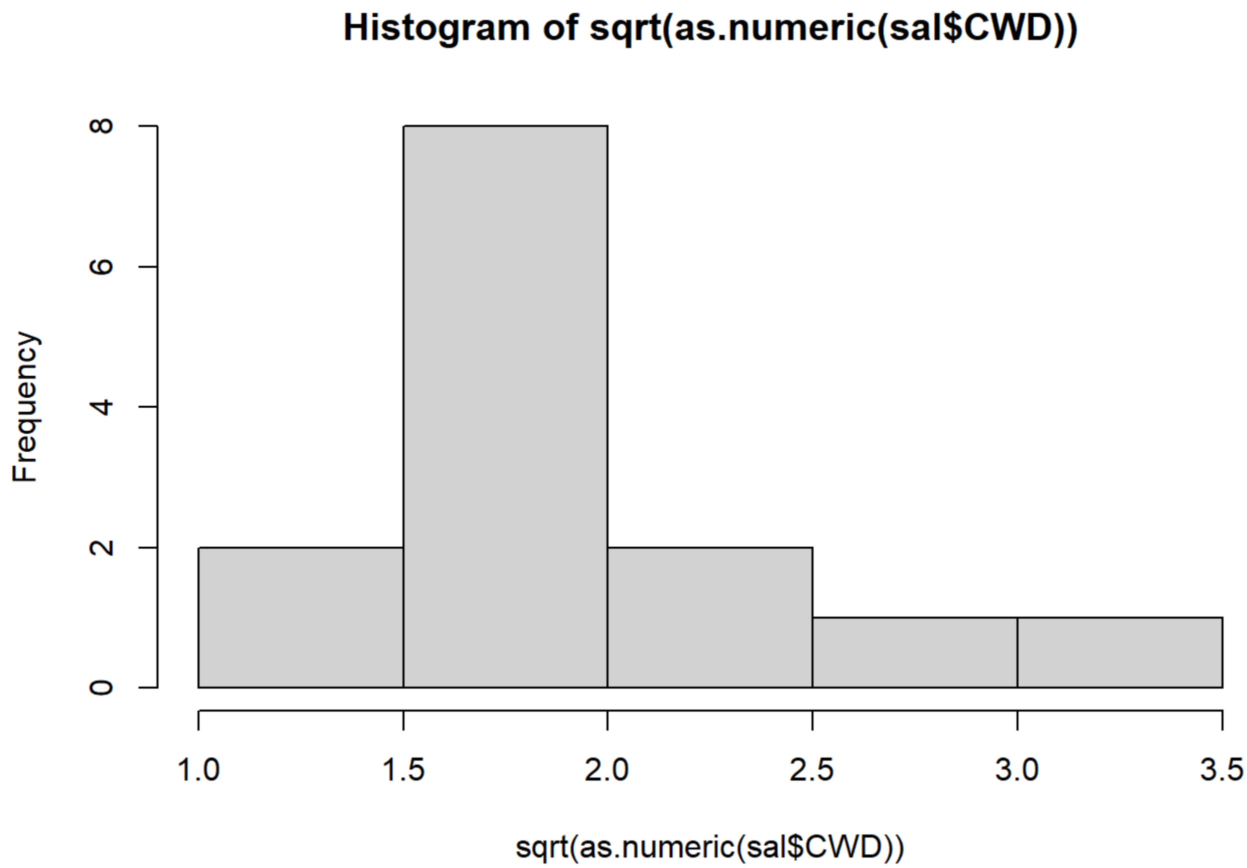
```r
shapiro.test(as.numeric(sal$CWD))
```

```
	Shapiro-Wilk normality test

data:  as.numeric(sal$CWD)
W = 0.66461, p-value = 0.0001627
```

```r
hist(log(as.numeric(sal$CWD)))
```




```r
hist(sqrt(as.numeric(sal$CWD)))
```




```r
shapiro.test(log(as.numeric(sal$CWD)))
```

```
	Shapiro-Wilk normality test

data:  log(as.numeric(sal$CWD))
W = 0.80104, p-value = 0.005141
```

```r
shapiro.test(sqrt(as.numeric(sal$CWD)))
```

```
	Shapiro-Wilk normality test

data:  sqrt(as.numeric(sal$CWD))
W = 0.73603, p-value = 0.0008956
```

Question 7

```r
sal_scaled <- scale(stat_vars)
```

```r
sal_scaled
```

```
         abund     fecund        CWD
2  -0.50509870 -0.7808331  0.2761127
3   0.32024557  1.2170226  1.3400332
4  -0.06444880  1.0285457  0.4179688
5   3.27539772  1.4808904  2.9004499
6   0.08593173  1.7447581 -0.2913116
7  -0.30925430 -0.7808331 -0.5395597
8  -0.47362371 -0.7431377 -0.5395597
9  -0.49810426 -0.7808331 -0.6104877
10 -0.39318762 -0.1777068 -0.4686316
11 -0.48761260 -0.7808331 -0.4331676
12 -0.46662927 -0.6300515 -0.4331676
13  0.49510664  0.7646779 -0.4331676
14 -0.47712093 -0.7808331 -0.5040957
15 -0.50160148 -0.7808331 -0.6814157
attr(,"scaled:center")
    abund    fecund       CWD
14.442857  2.071429  3.921429
attr(,"scaled:scale")
    abund    fecund       CWD
28.594128  2.652844  2.819760
```

```r
colMeans(sal_scaled)
```

```
        abund        fecund           CWD
-2.874685e-17 -7.137148e-17  8.326673e-17
```

No, because the rows represent different observations and not different variables. If it were to represent different variables, then one can use row standardization